🚀R&D DOCUMENT ON THE CREATION OF AN INTERNAL AND EXTERNAL LOAD BALANCER

PREPARED BY :
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PURPOSE :
To explore an Azure load balancer and create an internal and external load balancer, and verify if it's working

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🔎AZURE LOAD BALANCER OVERVIEW

- Azure Load Balancer is a cloud service that distributes incoming network traffic across backend virtual machines (VMs) or virtual machine scale sets (VMSS).

- Load balancing refers to the efficient distribution of incoming network traffic across a group of backend virtual machines.

- It operates at layer 4 of the OSI model.

- It serves as the single point of contact for clients.

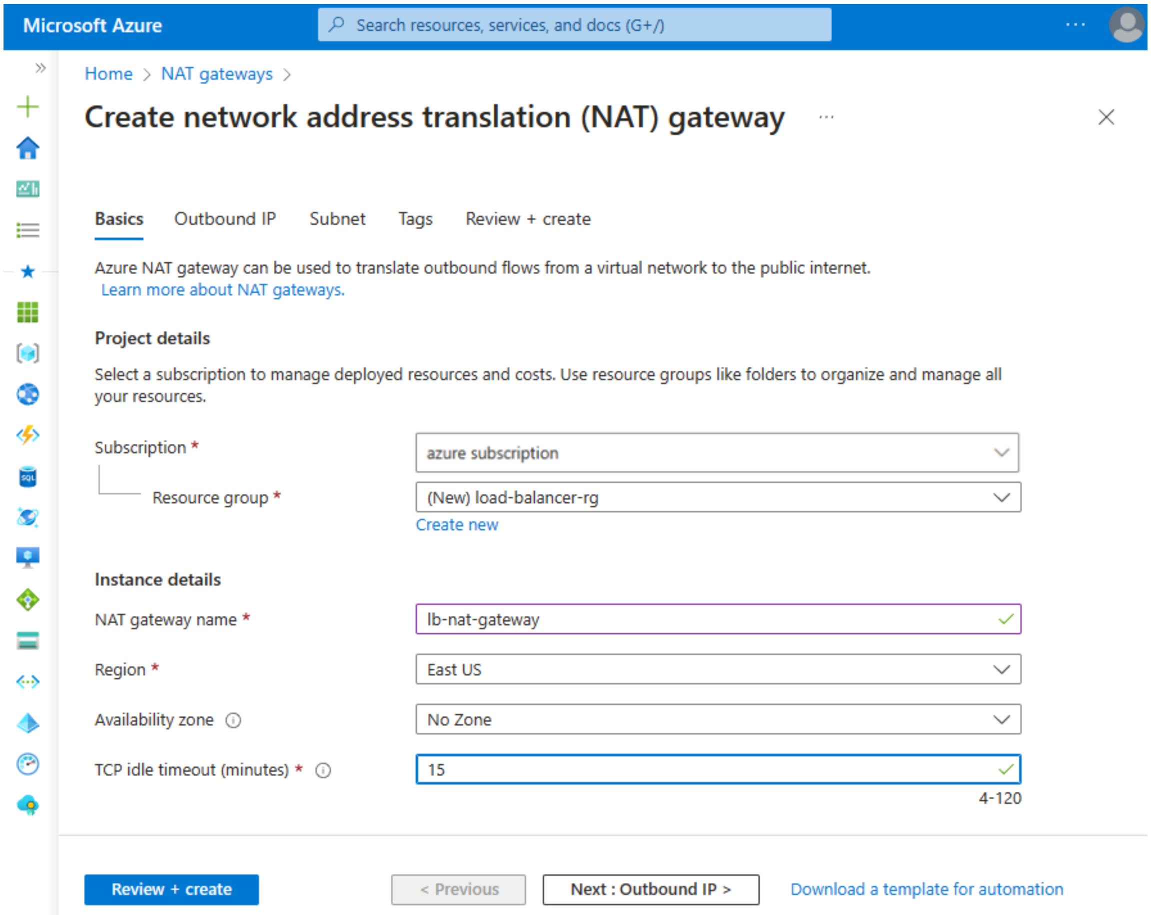
- Azure Load Balancer has three stock-keeping units (SKUs) - Basic, Standard, and Gateway. Each SKU is catered towards a specific scenario and has differences in scale, features, and pricing.

🔎CORE CAPABILITIES

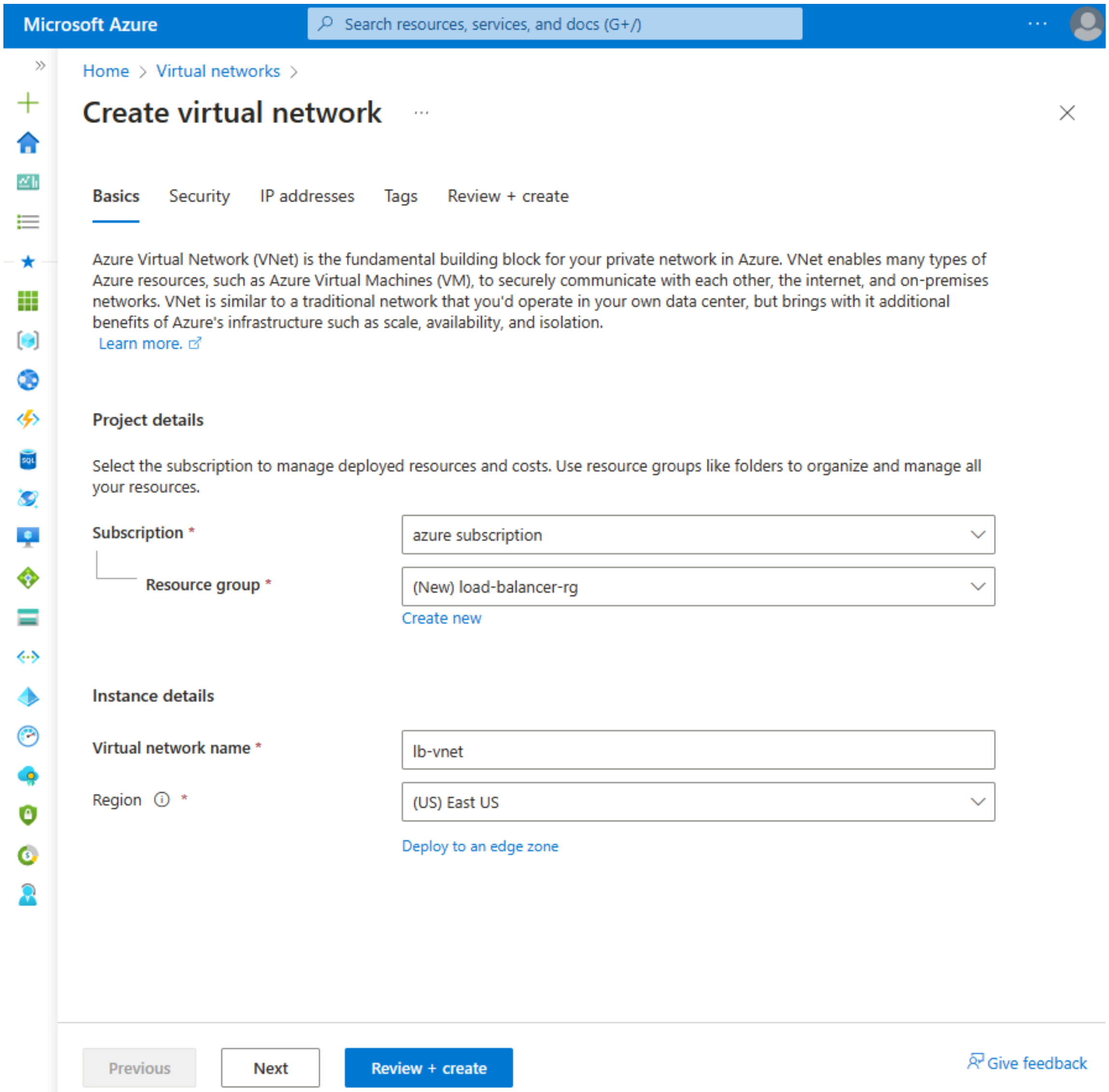
- High availability for the distribution of resources
- Scalability
- Low Latency by using pass-through load balancing
- Flexibility
- Health Monitoring

🔎DEPLOYMENT OF LOAD BALANCER - EXTERNAL

1. **Create a NAT Gateway**



2. Create a Virtual Network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Azure Virtual Network (VNet) is the fundamental building block for your private network in Azure. VNet enables many types of Azure resources, such as Azure Virtual Machines (VM), to securely communicate with each other, the internet, and on-premises networks. VNet is similar to a traditional network that you'd operate in your own data center, but brings with it additional benefits of Azure's infrastructure such as scale, availability, and isolation.
Learn more.
Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
azure subscription
Resource group *
(New) load-balancer-rg
Create new
Instance details
Virtual network name *
lb-vnet
Region *
(US) East US
Deploy to an edge zone
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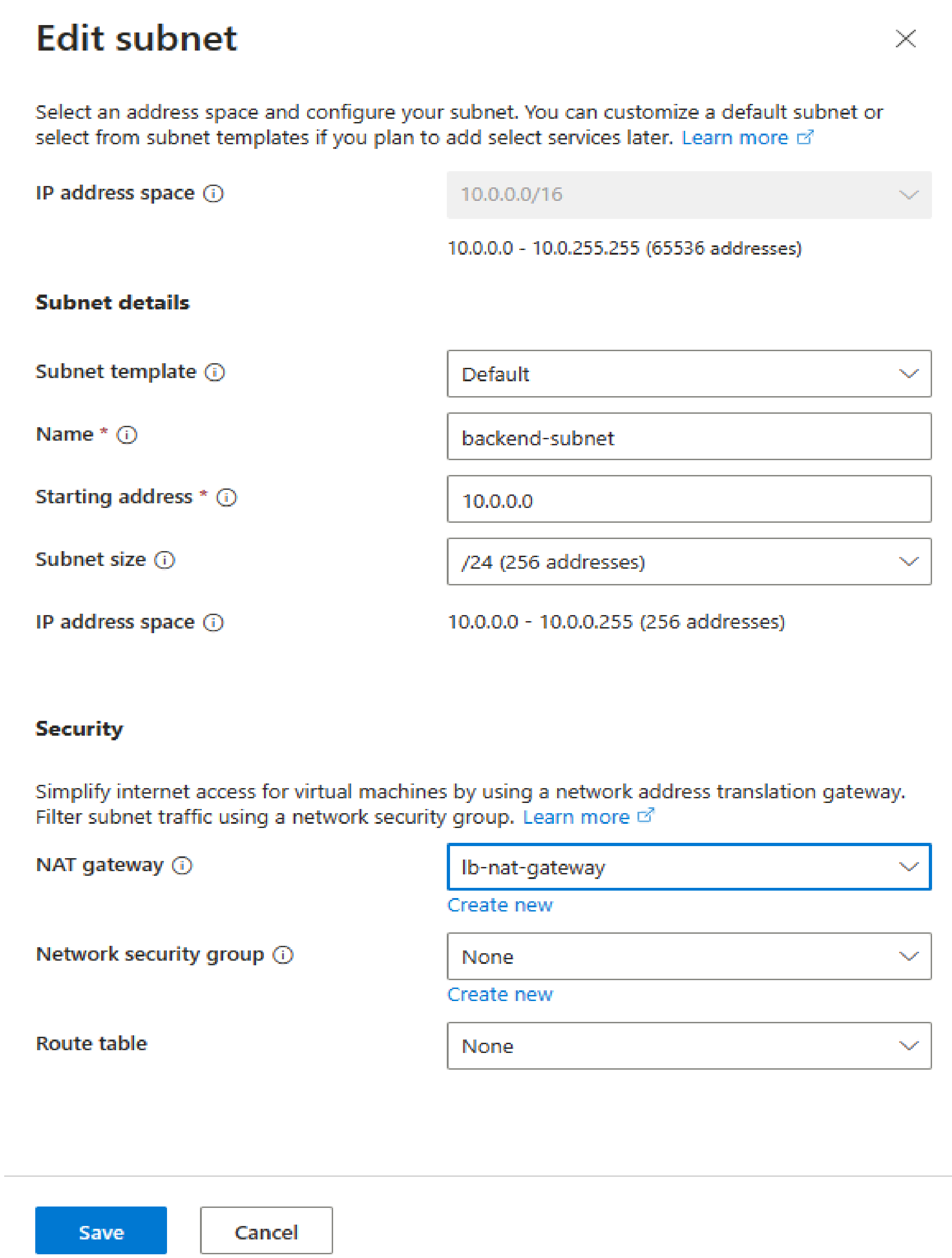
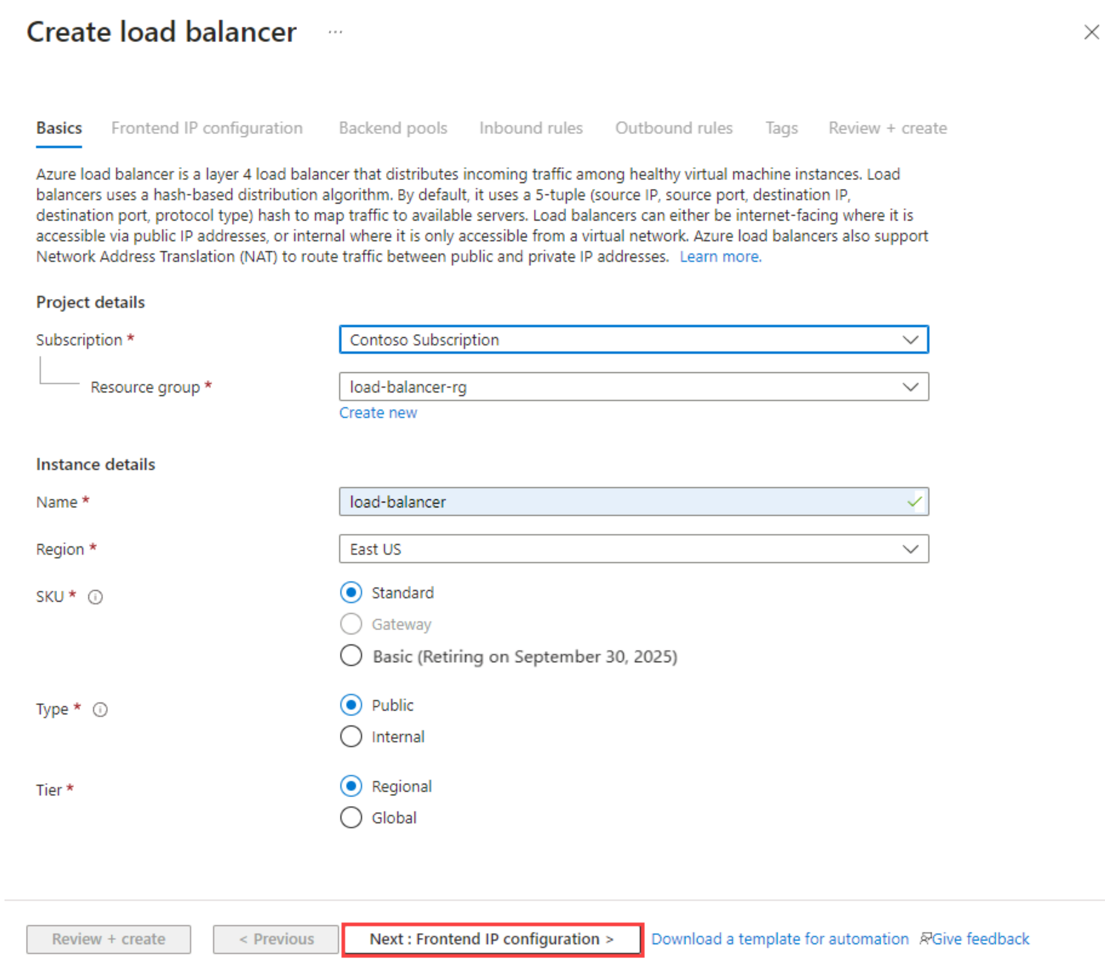


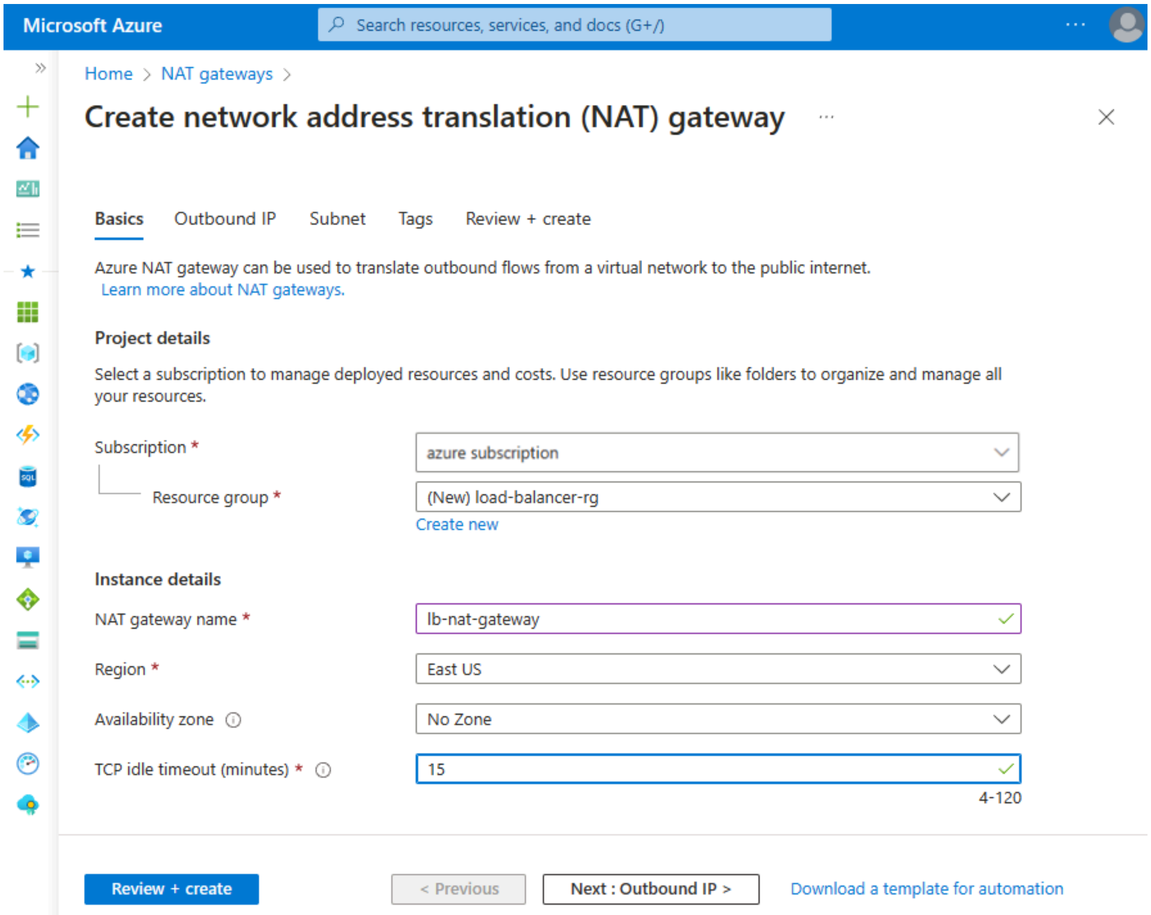
3. Create a Load Balancer



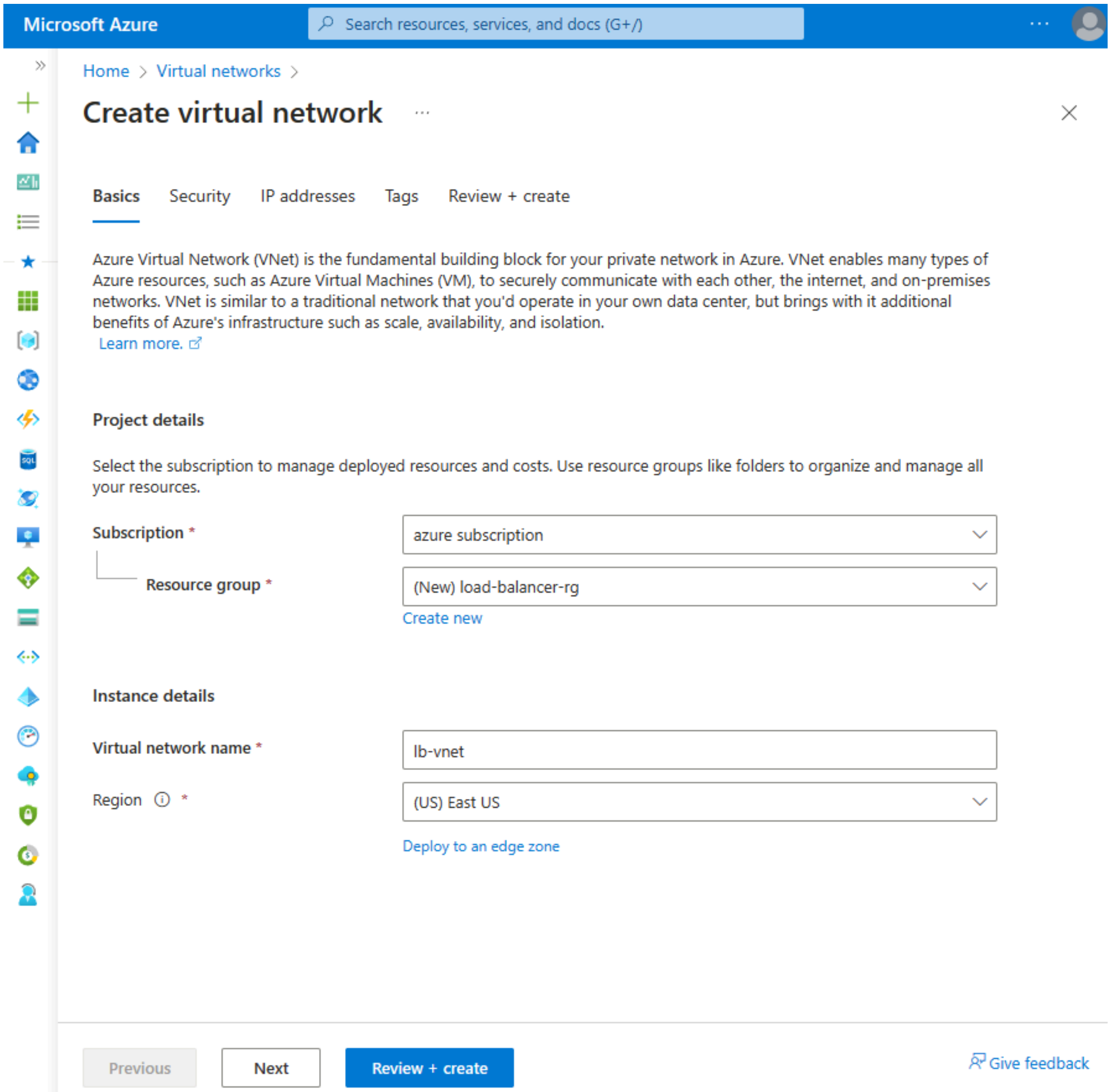
4. Testing the Load Balancer

🔎DEPLOYMENT OF LOAD BALANCER - INTERNAL

1. Create a NAT Gateway



2. Create a Virtual Network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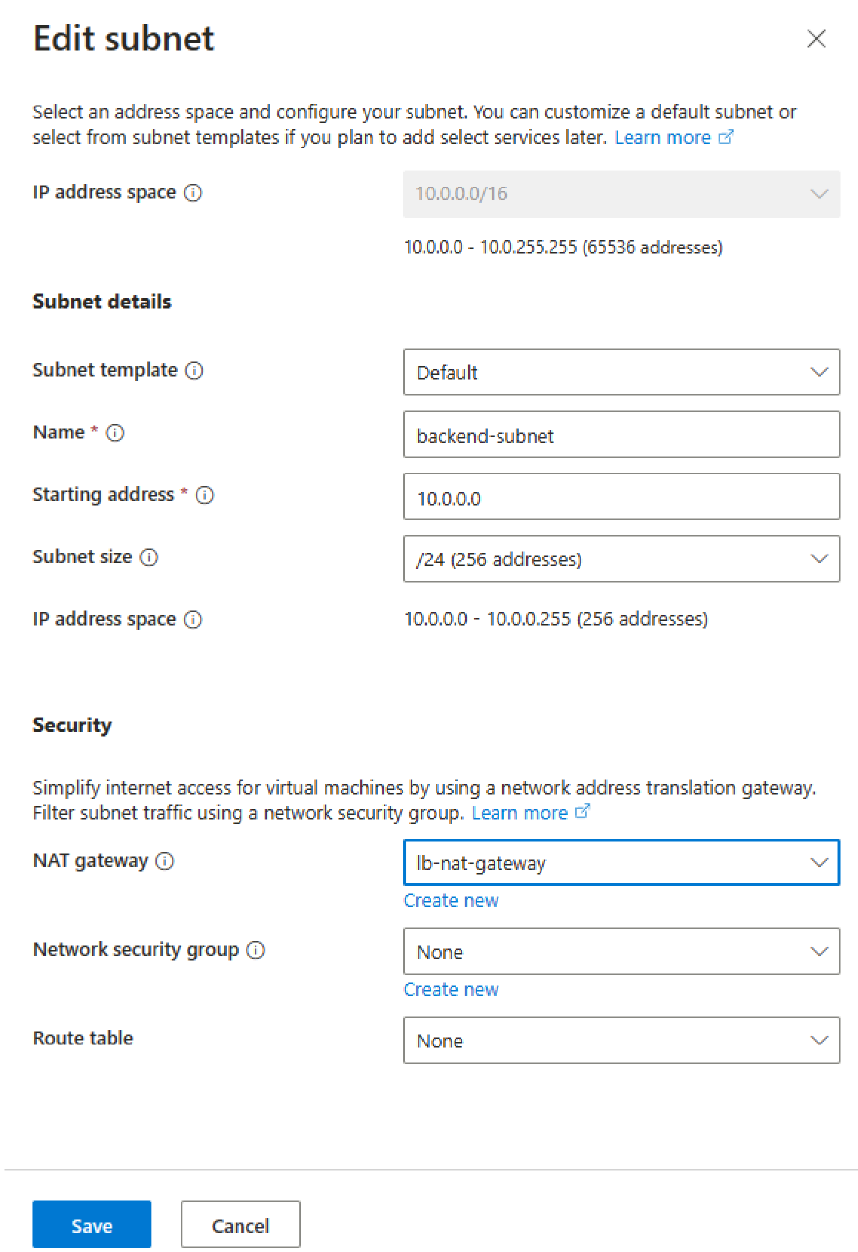

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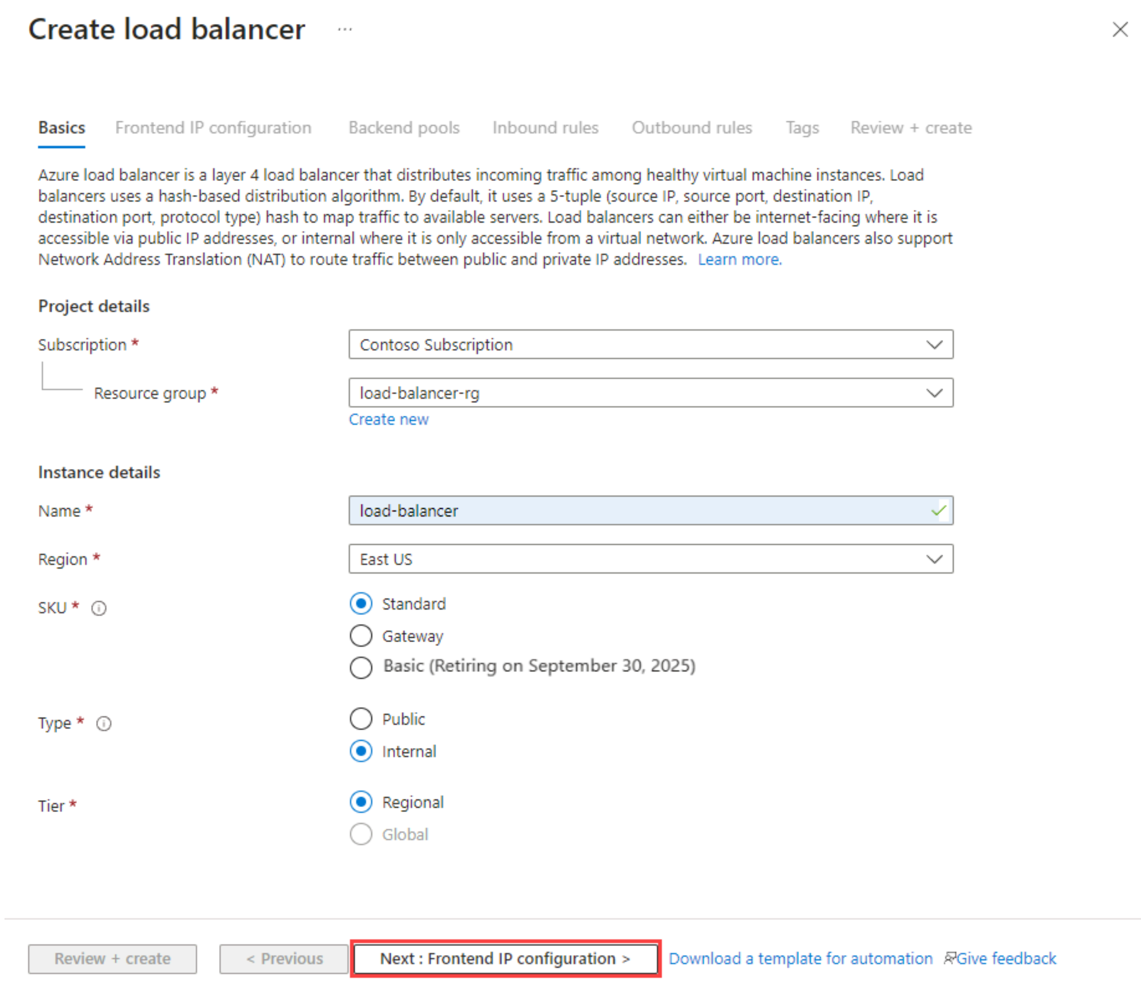
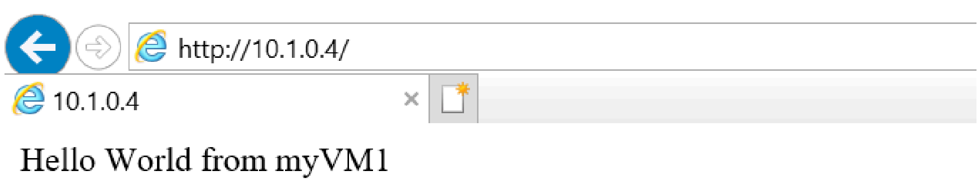


4. Test the Load Balancer



CONCLUSION

The deployment of internal and external Azure Load Balancers was successful. External Load Balancer ensured internet-facing traffic reached Web Tier VMs, while Internal Load Balancer securely managed traffic between App and DB tiers. Verification confirmed proper traffic distribution and tier isolation, achieving high availability and secure architecture.